XIANCE(NATE) XU

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EDUCATION

Northwestern University | M.S. in Electrical Engineering **Sep 2026 - Jun 2028 (Expected)**

University of California, Berkeley | Engineering Exchange - Digital Design (EECS 151), Analog IC Design (EE 140), Semiconductor Devices (EE 130) **Jan 2025 - May 2025**

Xi'an University of Technology | B.Eng. in Electronic Science and Technology | GPA: 3.6/4.0 **Sep 2022 - Jun 2026**

TECHNICAL SKILLS

HDL & Programming: Verilog, SystemVerilog, C/C++, Python

RTL / FPGA: RTL Design, FPGA Prototyping, AXI4-Lite, AXI-DMA, CDC, BRAM/FIFO, Memory-Mapped I/O, RISC-V, Zynq SoC

EDA / IC Design: Xilinx Vivado, Cadence Virtuoso, Mentor Graphics, DRC/LVS/PEX, Analog/Mixed-Signal IC Design

ENGINEERING EXPERIENCE

Xi'an Institute of Optics and Precision Mechanics, CAS | Research Intern **Jun 2025 - Sep 2025**

- Developed and verified a **16-bit ADC sampling interface** in Verilog; resolved clock-domain-crossing instability by redesigning two-stage synchronizers and handshake logic.
- Designed a DDR-less Zynq PL-PS data path using **AXI4-Lite control, BRAM/FIFO double buffering**, Verilog handshake logic, and a memory-mapped PS driver, enabling stable eight-channel acquisition at **50 MHz**.
- Optimized burst length and sampling windows to achieve **145 MB/s measured throughput**; the architecture was adopted as the lab's standard interface for subsequent high-speed acquisition projects.

Xidian University | Research Assistant, Institute of Embedded Computing Technology **Jul 2024 - Aug 2024**

- Deployed a MobileNetV2 four-class image classifier to a **Zynq-7000 SoC**, integrating AXI-DMA and PL-side buffering for **12 FPS** inference at 320x240; trained on 2,000 images to **82% test accuracy**.

HARDWARE PROJECTS

Five-Stage RISC-V Processor | EECS 151, UC Berkeley **Mar 2025 - May 2025**

- Led the design and FPGA implementation of a **32-bit five-stage RISC-V CPU** with CSR registers and hardware interrupt handling.
- Achieved **95 MHz Fmax** and **1.24 CPI** with a non-forwarding pipeline; simplified hazard and control logic using stall/NOP insertion to reduce critical-path complexity and improve timing closure.
- Reduced LUT use by **22% (1,293 to 1,008)** and saved **34 BRAM blocks**; debugged pipeline hazards and latch inference using assertions and timing reports, and verified functionality through RTL simulation, assembly-level tests, ISA validation, and FPGA prototyping.

Mixed-Signal IoT IC Design & Tape-Out | EE 140, UC Berkeley **Apr 2025 - May 2025**

- Led the design and tape-out of a mixed-signal IoT sensor IC integrating an **8-bit SAR ADC, PGA, BGT, analog MUX, and LDO**, supporting **1.6-3.2 V** operation and **0-70 °C**.
- Designed and verified folded-cascode and two-stage amplifiers in **Cadence Virtuoso**; completed **DRC/LVS** verification and parasitic extraction (**PEX**), followed by AC/transient post-layout simulations.
- Built AREG/DREG LDOs regulating 1.6-3.2 V to **1.8 V**; achieved **3.7 mV** BGT reference error and mitigated output oscillation with capacitive feedback and unity-gain buffers.
- Integrated a **StrongARM latch comparator** into the SAR ADC signal path and verified functionality across **PVT corners** and post-layout extracted simulations.

EXPERIENCE & HONORS

CloudSublet | Founder & Engineering Lead **May 2026 - Present**

- Built and launched a student-housing platform serving four university communities using JavaScript, Node.js, WeChat Mini Program, and Tencent CloudBase; led product development through an **80-user beta**.

Honors & Leadership: College-Level One-Star Scholarship (2025); University Outstanding Student Scholarship (2023); University Advanced Individual in Scientific and Technological Innovation (2023); Video Team Lead, XUT - grew the official Douyin account by 20K followers.